PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH
BADJI MOKHTAR UNIVERSITY -ANNABA-
FACULTY OF MEDICINE
DEPARTMENT OF MEDICINE
LABORATORY OF MEDICAL AND SURGICAL ANATOMY

Anatomical language

Course notes intended for first-year medical students.

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ANATOMICAL LANGUAGE

Course outline

Introduction

1. Anatomical language
2. Dynamic benchmarks
3. Location and direction terms
4. Structural organization of the human body
5. Topographical organization of the human body

Conclusion

Bibliography

Introduction

Anatomical language, or anatomical nomenclature, is a International Nomenclature of Human Anatomy, written in Latin, the result of an international agreement, aims to universalize the study of the human body.

1. Anatomical language

1.1. Nomina Anatomica (NA)

For a "universal" study of the human body, an international agreement (adopted in Paris in 1955) developed an international anatomical nomenclature in Latin, the Nomina Anatomica: NA

This nomenclature relates to a well-defined position of the human body, where 3 axes are distinguished, which allow us to define 3 planes, and reference points which designate the direction of movements.

1.2. Reference anatomical position

All anatomical descriptions refer to a "Reference Anatomical Position", where the subject is:

- A living, upright human body.
- The upper limbs hang down along the trunk.
- Palms facing forward.
- The gaze should be straight and horizontal.
- With legs extended,
- Feet together, placed on the ground

1.3. Reference anatomical axes

1.3.1. The body's axis: It is the vertical line dropped from a point located at the top of the skull that passes through the center of gravity of the body (located at the level of the pelvis).

1.3.2. The axis of the hand : corresponds to the longitudinal axis that passes through the 3rd finger.

1.3.3. The axis of the foot : corresponds to the longitudinal axis that passes through the 2nd toe.

1.4. Reference plans

Three perpendicular planes:

- **The frontal (coronal) plane:**vertical plane parallel to the forehead, which divides the body into 2 parts:
  - Previous.
  - Posterior.
- **The sagittal plane:**vertical plane, anteroposterior. It can be:
  - Median sagittal: passes through the midline and divides the body into 2 symmetrical parts, right and left.
  - Sagittal paramedian: passes beside the midline and divides the body into 2 parts, internal and external.
- **The transverse plane (horizontal or axial):**a plane parallel to the ground, which divides the body into 2 parts:
  - Superior.
  - Lower.

2. Dynamic benchmarks

Body movements occur in a plane, and in relation to an axis.

- **Along the transverse axis**we have:

Flexion: brings a part of the body forward

Extension: brings a part of the body backwards.

- **Along the vertical axis**we have:

Internal rotation: rotary movement from outside to inside.

External rotation: rotary movement from the inside out.

- **Along the sagittal axis**we have:

Adduction: brings a part of the body closer to the midline.
Abduction: moves a part of the body away from the midline.

3. Location and direction terms

- **Medial:**near the median sagittal plane.
- **Lateral:**far from the median sagittal plane.
- **Superior, cranial, or cephalic:**near the upper end.
- **Inferior or caudal:**near the lower end.
- **Proximal:**close to the root of the limb.
- **Distal:**far from the root of the limb.
- **Right :**closer to the right.
- **LEFT :**closer to the left.
- **Ipsilateral:**belong to the same side.
- **Contralateral:**belong to two different sides.
- **Anterior or ventral:**located near the anterior wall.
- **Posterior or dorsal:**located near the posterior wall.
- **Internal :**look inside a cavity or organ.
- **External :**look at the outside of a cavity or organ.
- **Superficial:**close to the surface.
- **Deep :**far from the surface.

4. Structural organization of the human body

4.1. Levels of structural organization

- **The chemical level**: it is the set of chemical substances necessary for the maintenance of life.
- **The cellular level**: the cell is the basic structural and functional unit.
  Ex: neuron, blood cell.
- **The tissue level**: tissue is a group of cells that play a homologous role.
  Example: bone tissue.
- **The organic level**: the organ brings together different tissues for a specific function.
  Examples: liver, brain.
- **The systemic level**: the system is a set of organs that are comparable from a structural point of view.
  Ex: skeletal system, central nervous system.
- **The systemic level**: the apparatus is the set of dissimilar organs which participate in the same function.
  Examples: digestive system, cardiovascular system.

4.2. Organ systems

4.2.1. Devices of relational life

4.2.1.1.The locomotor system: encompasses the skeletal, articular and muscular systems.
4.2.1.2. The nervous system: includes the central, peripheral and autonomic nervous systems.
4.2.1.3. The sensory apparatus: includes the 5 sense organs (touch, taste, smell, sight and hearing).

4.2.2. Nutrition devices

4.2.2.1.The digestive system:includes the digestive tract and associated glands.
4.2.2.2.The circulatory system:includes the heart and the arterial, venous and lymphatic systems.
4.2.2.3.The respiratory system:includes the respiratory tract and lungs.
4.2.2.4.The urinary system:includes the kidneys and urinary tract.
4.2.2.5.The endocrine system:represented by the endocrine glands.

4.2.3. Generating (or reproducing) apparatus

4.2.3.1.Male genitalia: includes the testicles, prostate, and spermatic ducts
4.2.3.2.Female genitalia: includes the ovaries and the genital tract.

5. Topographical organization of the human body

5.1.Human body shape

The shape of the human body is determined by that of the osteo-cartilaginous skeleton. Muscles are added to fill and reinforce the walls.

5.2. The head

It carries the sensory apparatus. It contains the initial segments of the digestive and respiratory systems, and it contains a large part of the nervous system.

5.3.The neck

It supports the head. It is a passageway for visceral, vascular, and nervous elements.

5.4.The trunk

Contains the major cavities of the body: The thoracic cavity, at the top, and the abdominopelvic cavity, at the bottom.

5.5.The members

The limbs are extensions of the trunk, formed essentially by bones, joints, muscles, vessels and nerves.

CONCLUSION

Anatomy is an ancient and fundamental science, essential for understanding physiology and mastering radiology and surgery. Effective and uncomplicated surgical interventions are based on a thorough understanding of anatomy.

Bibliography

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3. Vincent Delmas,Dominique Brémond-GignacGeneral Anatomy.Elsevier Masson, 2008 - 323 pages

Langage Anatomique



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Plan du cours

- Nomenclature anatomique
- Position anatomique de référence
- Axes de référence
- Plans de référence
- Repères dynamiques
- Termes de localisation et direction
- Organisation structurale du corps humain
- Organisation topographique du corps humain







Nomenclature anatomique



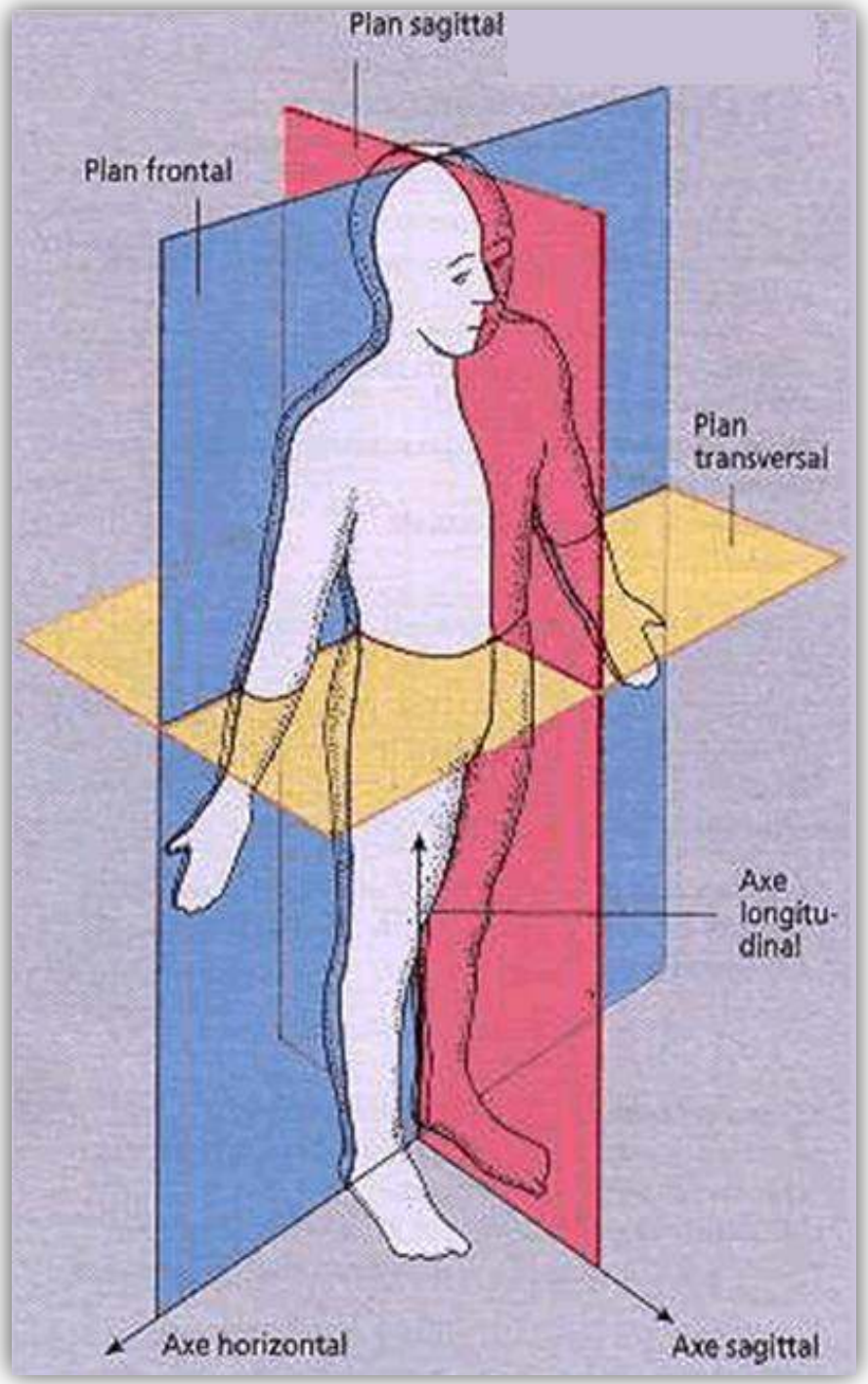


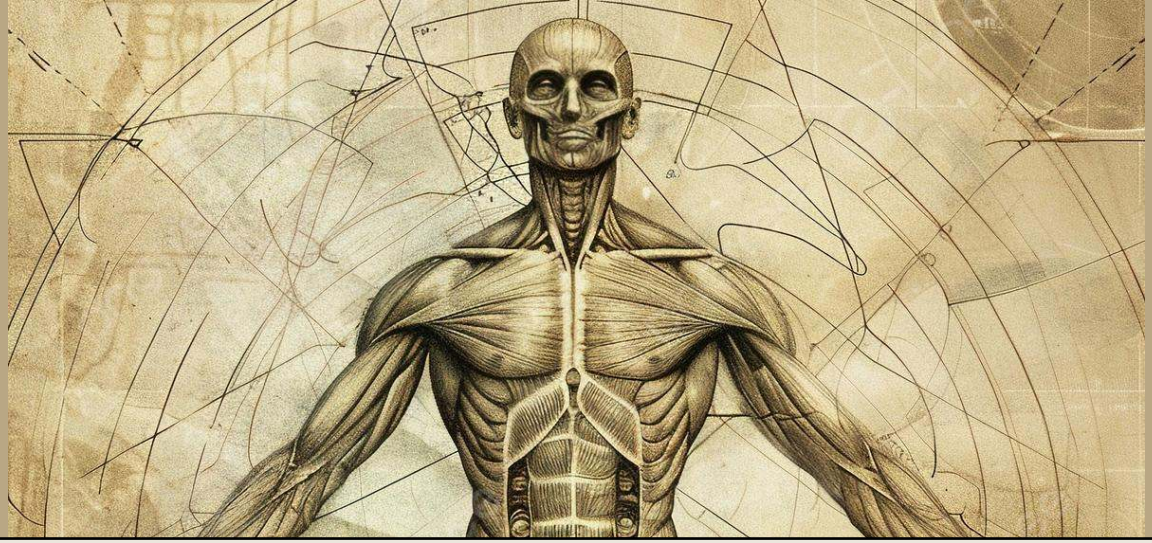
- **La nomenclature anatomique est une** Nomenclature **Internationale** de l'anatomie humaine, rédigée en **Latin**, le fruit d'un accord international, le but est d'universaliser l'étude du corps humain.
  - **La nomina anatomica de Bâle** publiée en 1895.
  - **La nomina anatomica (NA)**, adoptée à **Paris** publiée officiellement en 1956.
  - **Terminologia Anatomica (TA)** publiée en 1998, adoptée dans tous les pays.

La nomina anatomica ou Terminologia anatomica se rapporte à une **position bien définie** du corps humain.

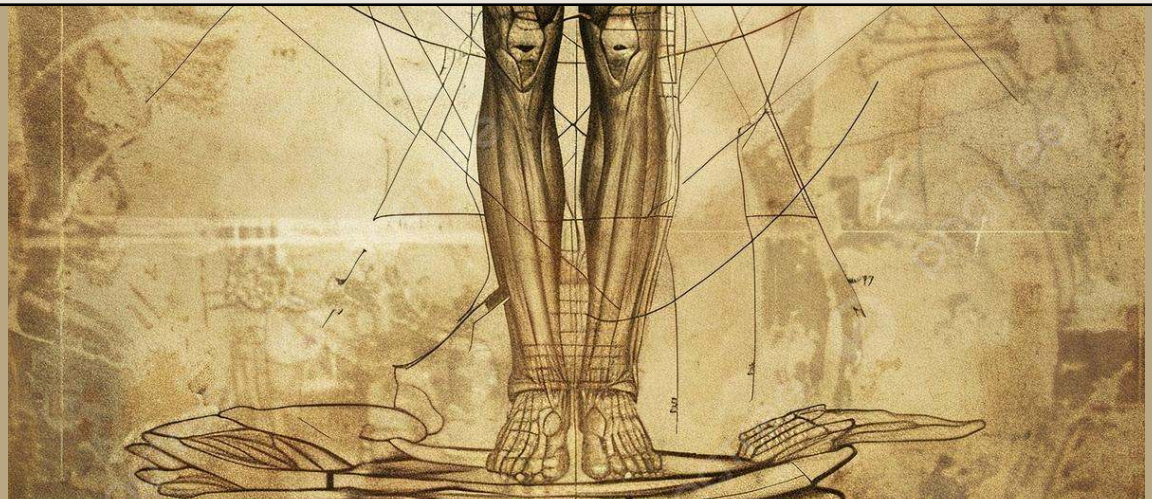
Où on distingue:

- **3 axes**
- **3 plans**
- **Repères dynamiques**



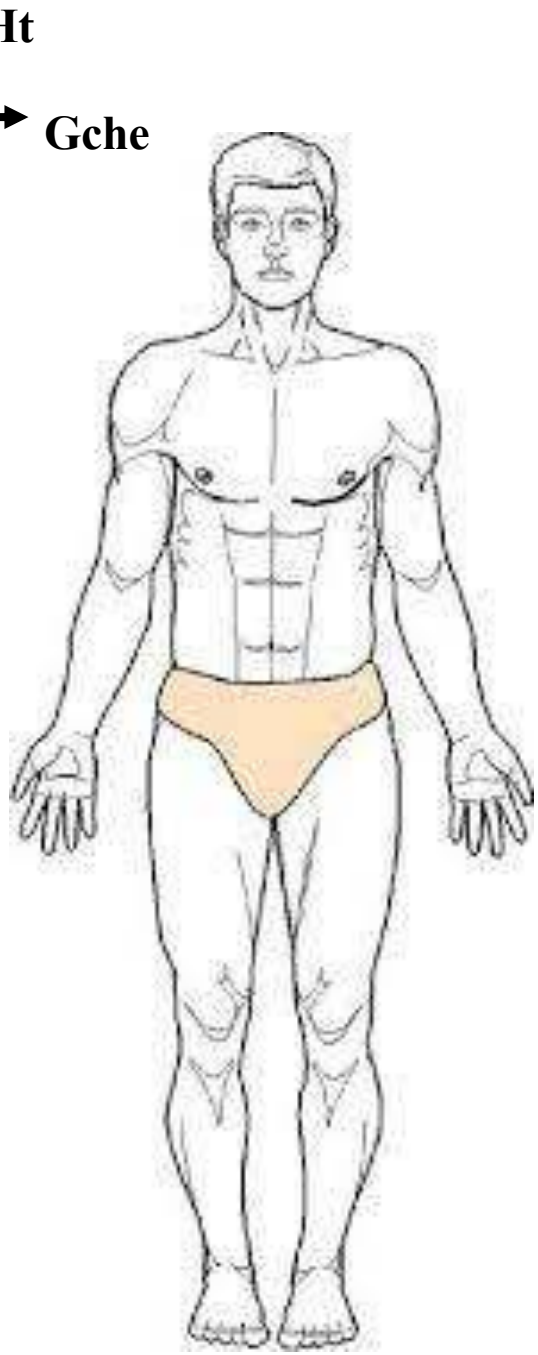


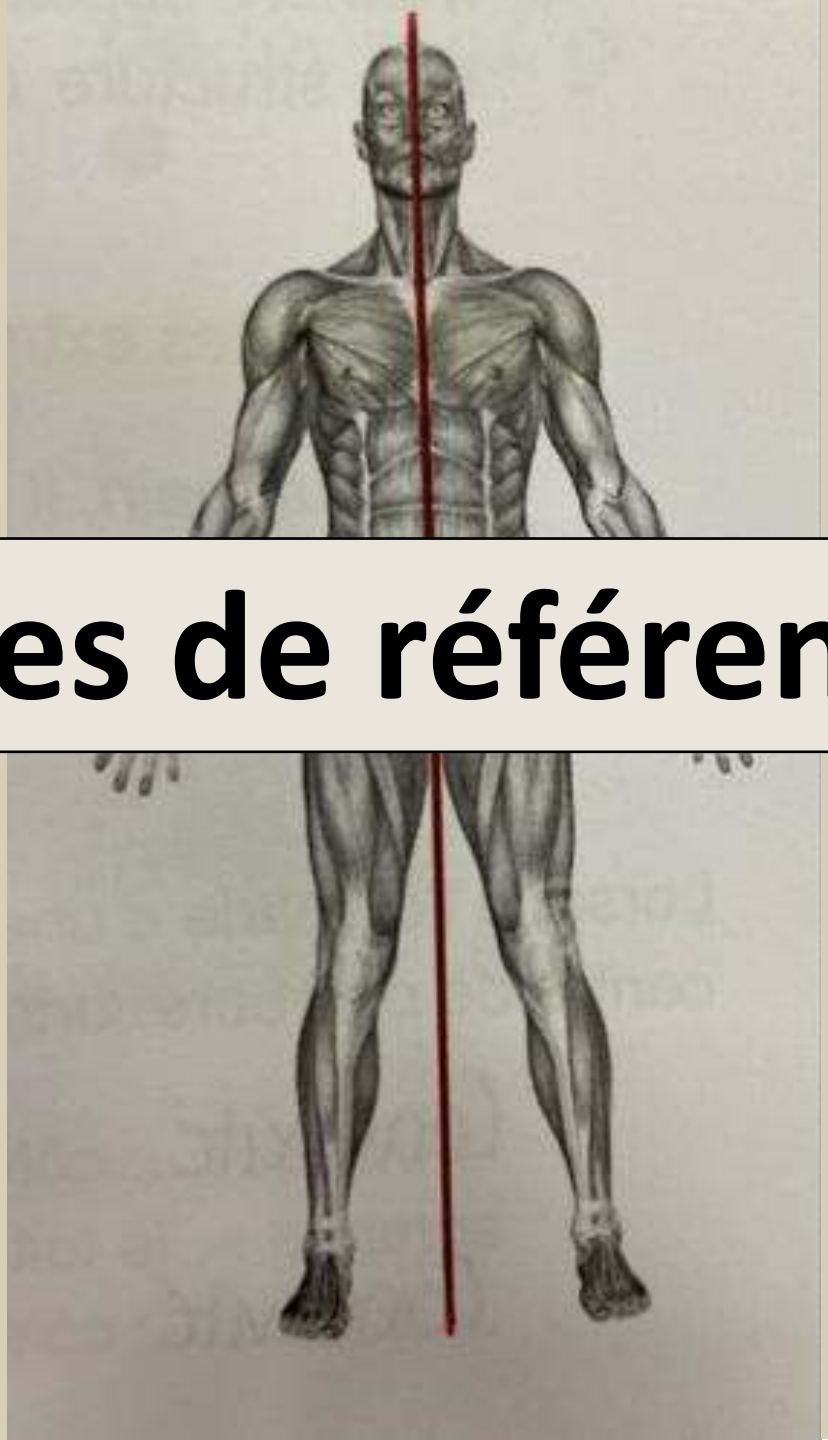
Position anatomique de référence



La position anatomique de référence sur laquelle se basent **toutes les descriptions anatomiques**, elle est définie par une convention internationale ainsi :

- **Corps humain, vivant** et **debout,**
- **les membres supérieurs** pendants le long du corps,
- **la paume des mains** tournée en avant,
- **le regard** droit et horizontal,
- **Les jambes** étendues,
- **Les pieds** joints posés sur sol.

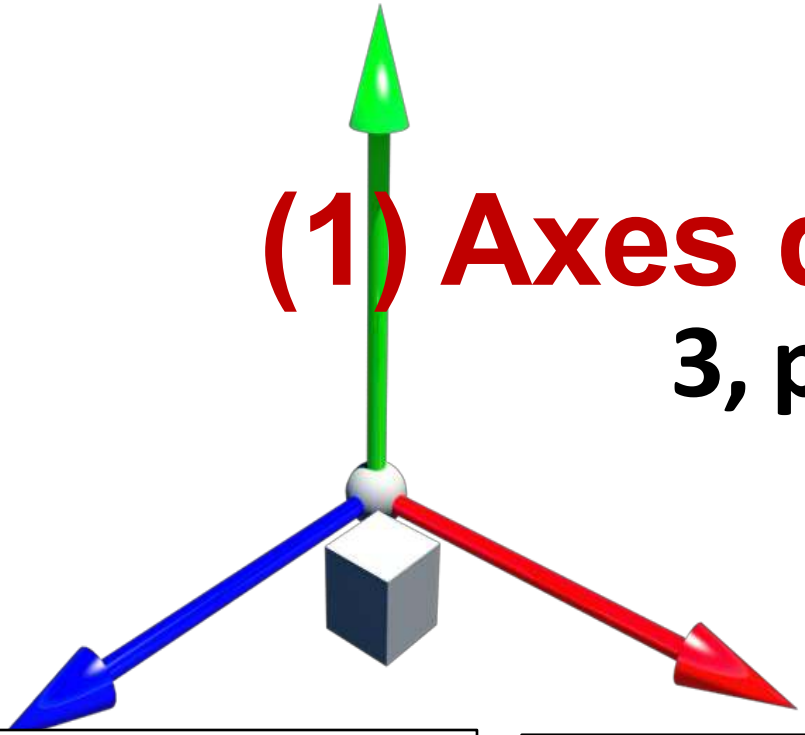




Axes de référence

Axe vertical (longitudinal):
orienté **du bas en haut**

(1) Axes de l'espace:
3, perpendiculaires

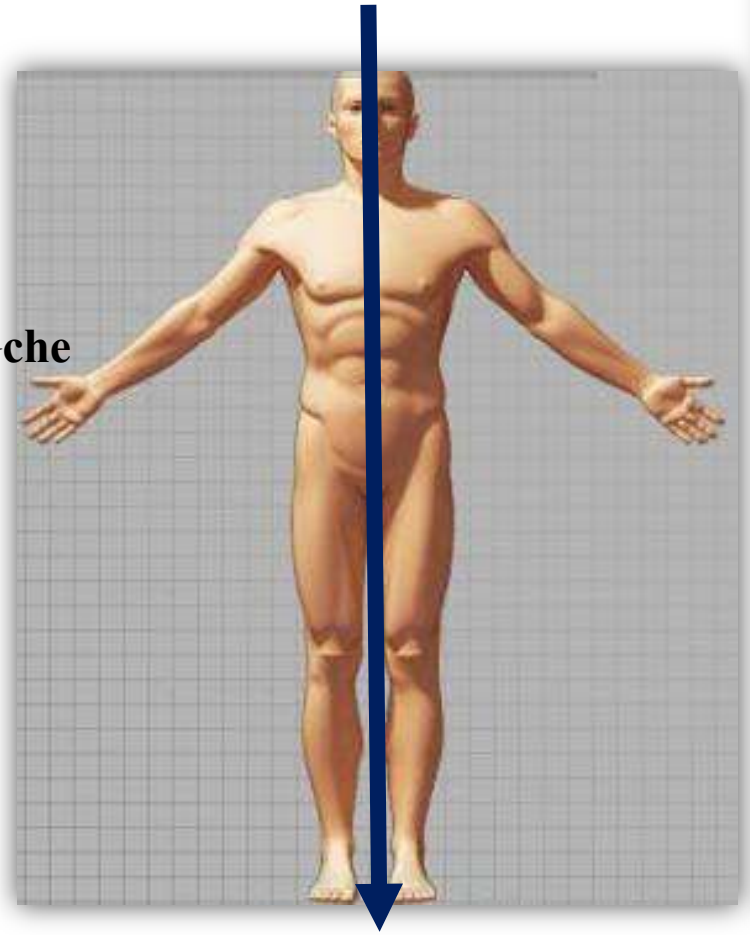


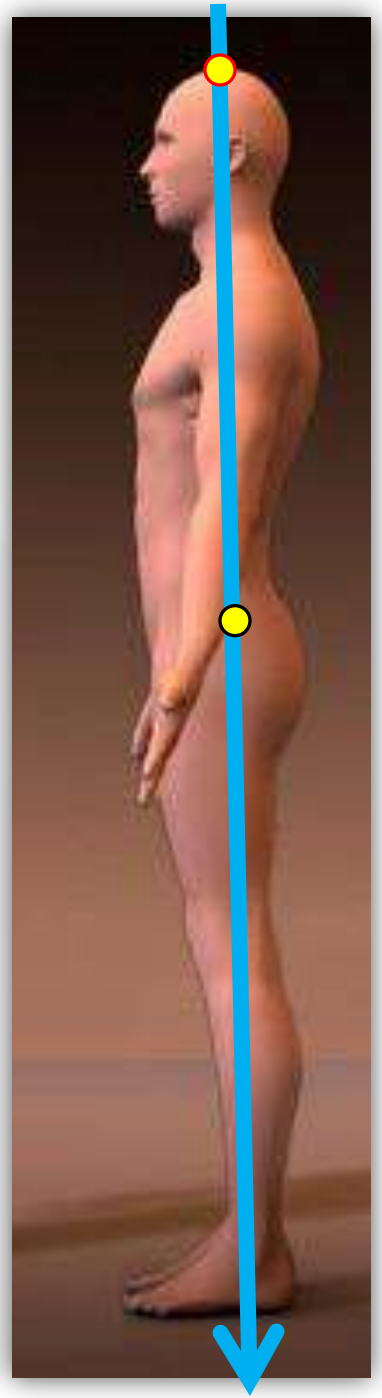
Axe sagittal:
orienté **d'arrière en avant**

Axe transversal (horizontal):
orienté **de droit à gauche**

(2) Axe du corps

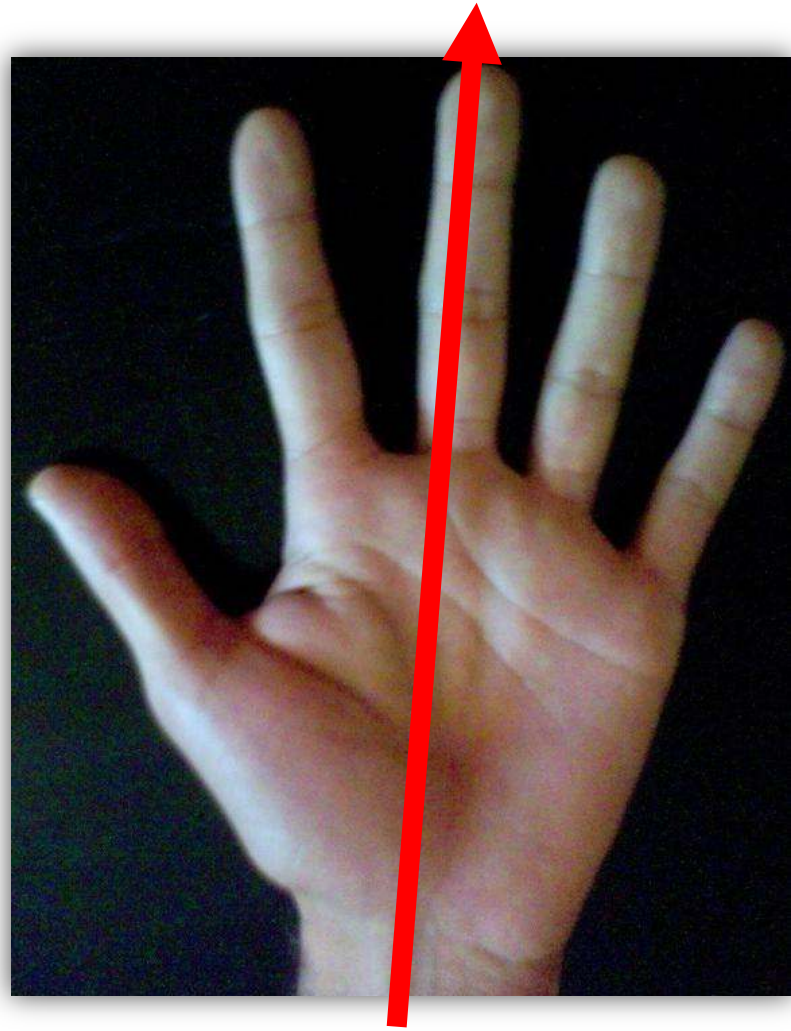
Une ligne verticale abaissée du **vertex** et passe par **le centre de gravité du corps** (au niveau du **pelvis**).





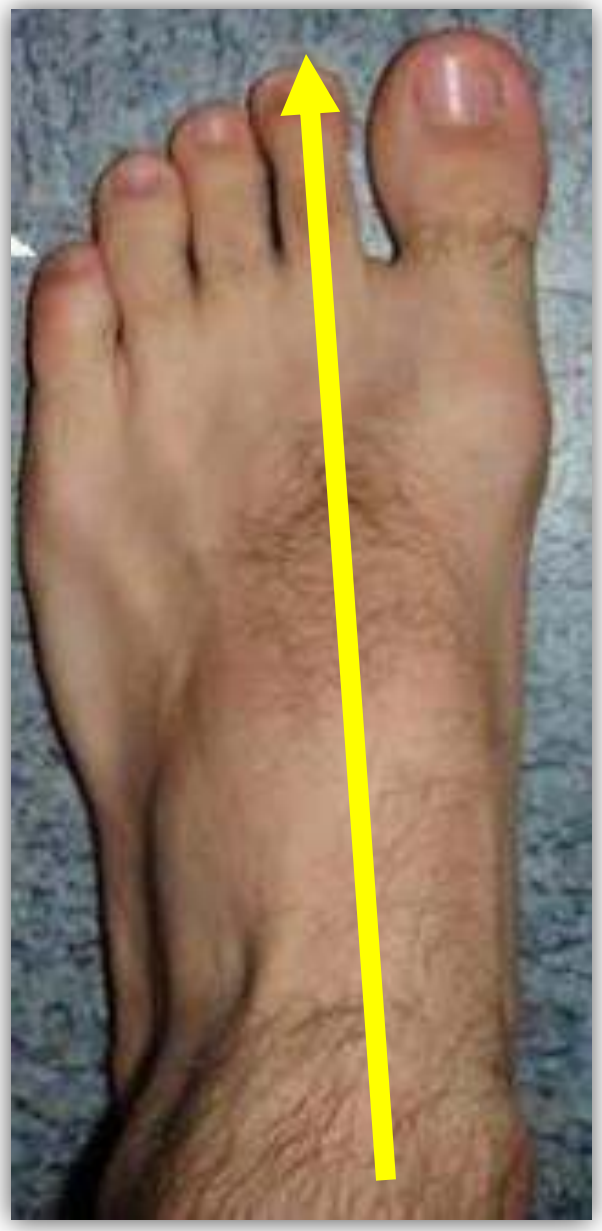
(3) Axe de la main

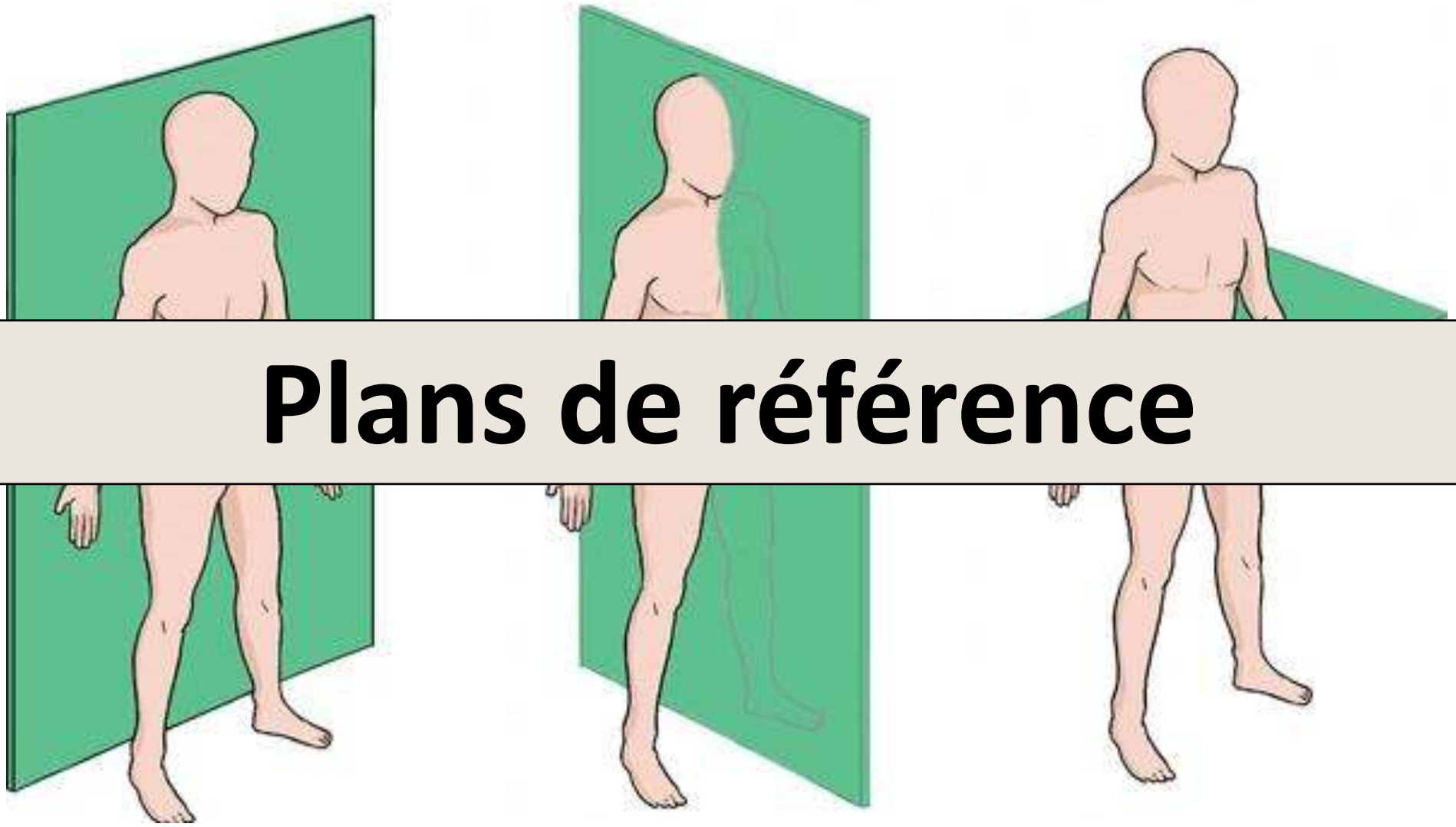
Axe longitudinal qui passe par le **3ème doigt.**



(4) Axe du pied

Axe longitudinal qui passe par le **2ème orteil.**





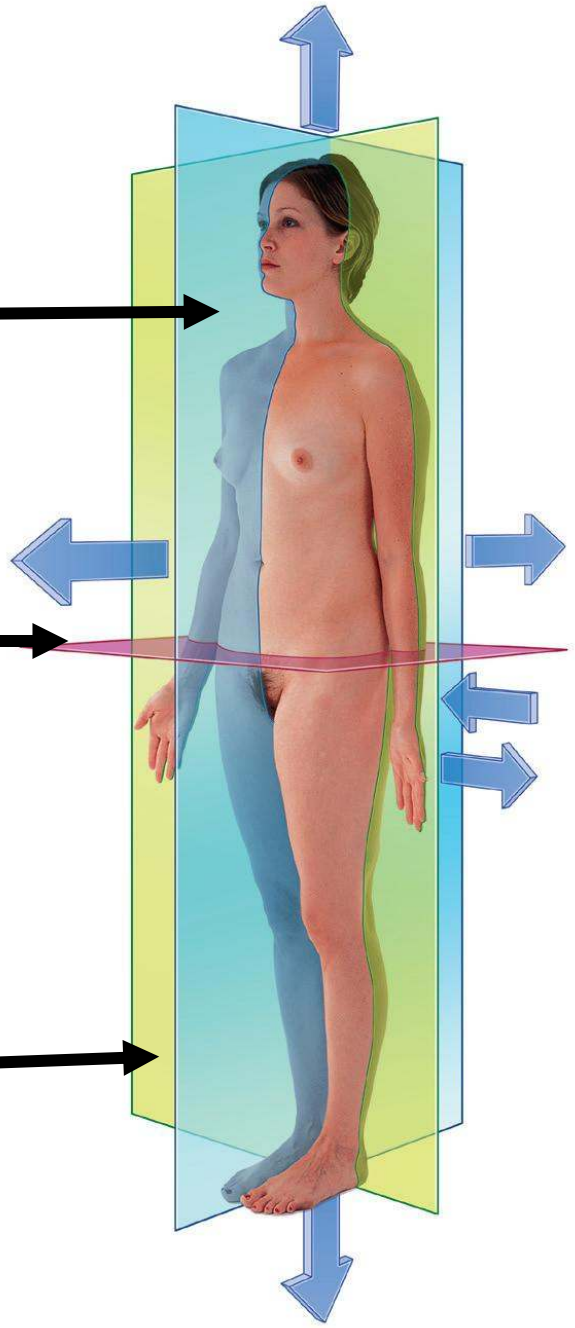
Plans de référence

3 Plans perpendiculaires

Plan sagittal

Plan transversal
(horizontal ou axial)

Plan frontal (coronal)

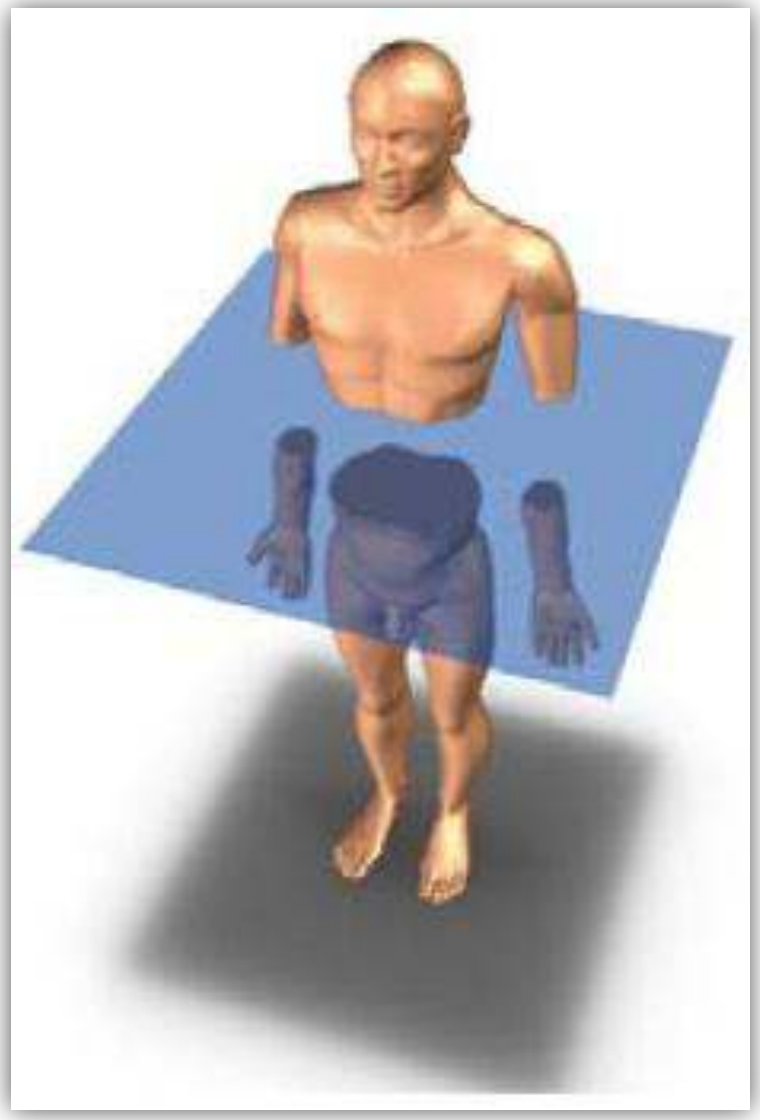


Plan transversal

Parallèle au sol

Divise le corps en **2 parties**:

- **Supérieure (crâniale)**
- **Inférieure (caudale)**



Plan frontal

Vertical, **parallèle au front**

Divise le corps en **2 parties**:

- **Antérieure**
- **Postérieure**



Plan sagittal



Vertical, antéro-postérieur

Divise le corps en **2 parties**:

- **Soit symétriques, droite** et **gauche** (le **plan sagittal médian**)
- **Soit asymétrique, médiale** et **latérale** (le **plan sagittal paramédian**)

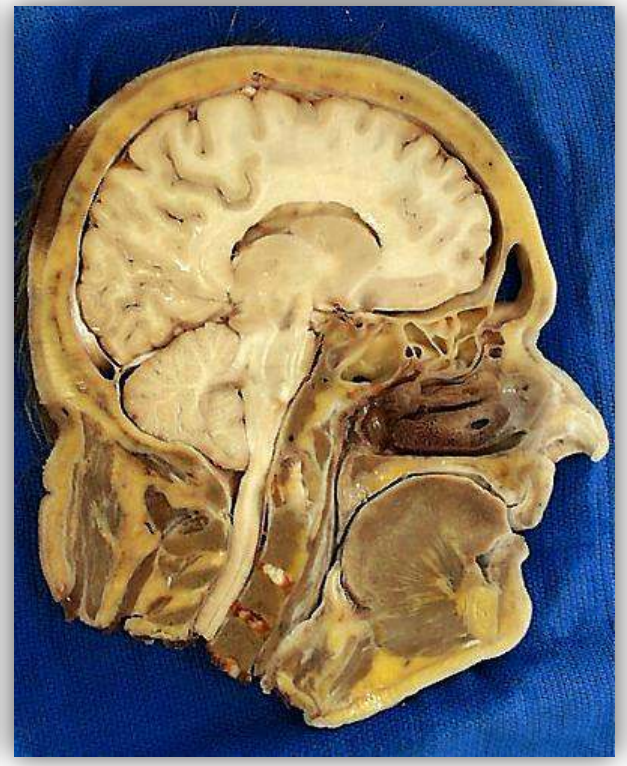




















Repères dynamiques





Les repères dynamiques désignent **la direction des mouvements**

Chaque mouvement se fait:

- dans un **plan**
- par rapport à un **axe**



(1) Flexion :

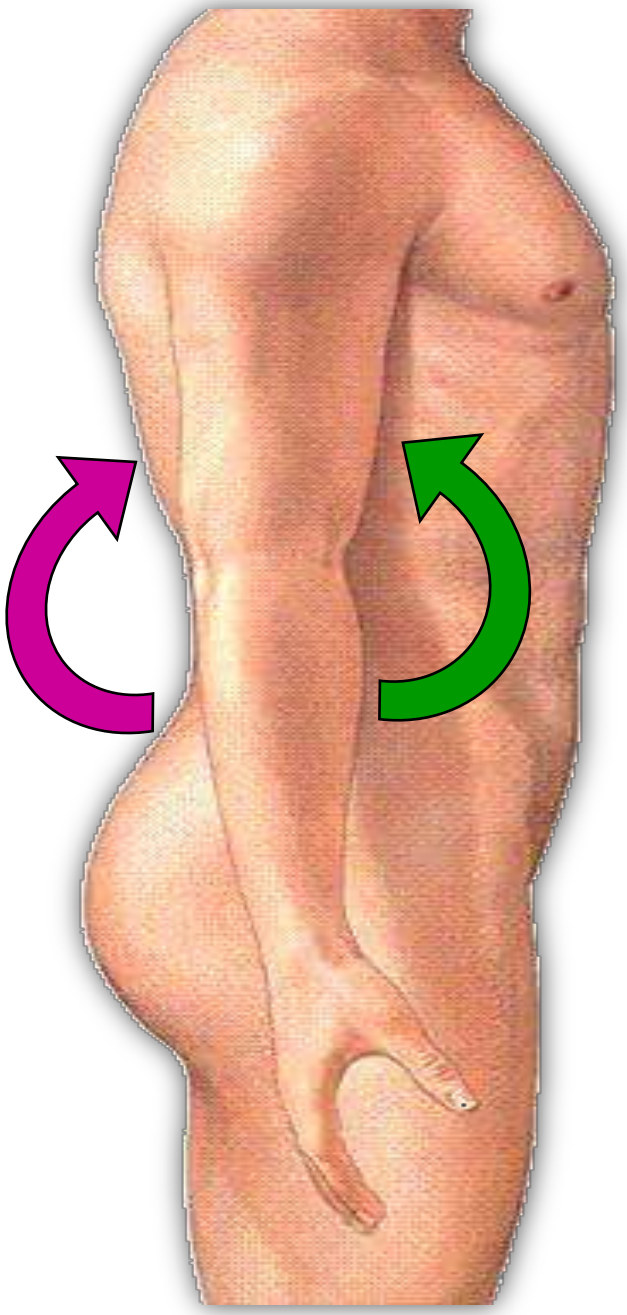
ramène une partie du corps vers **l'avant**

(2) Extension :

ramène une partie du corps vers **l'arrière**

Se font :

- Selon un **axe transversal**
- Dans un **plan sagittal**



(3) Rotation médiale:

mouvement **rotatoire** en allant **de dehors en dedans**

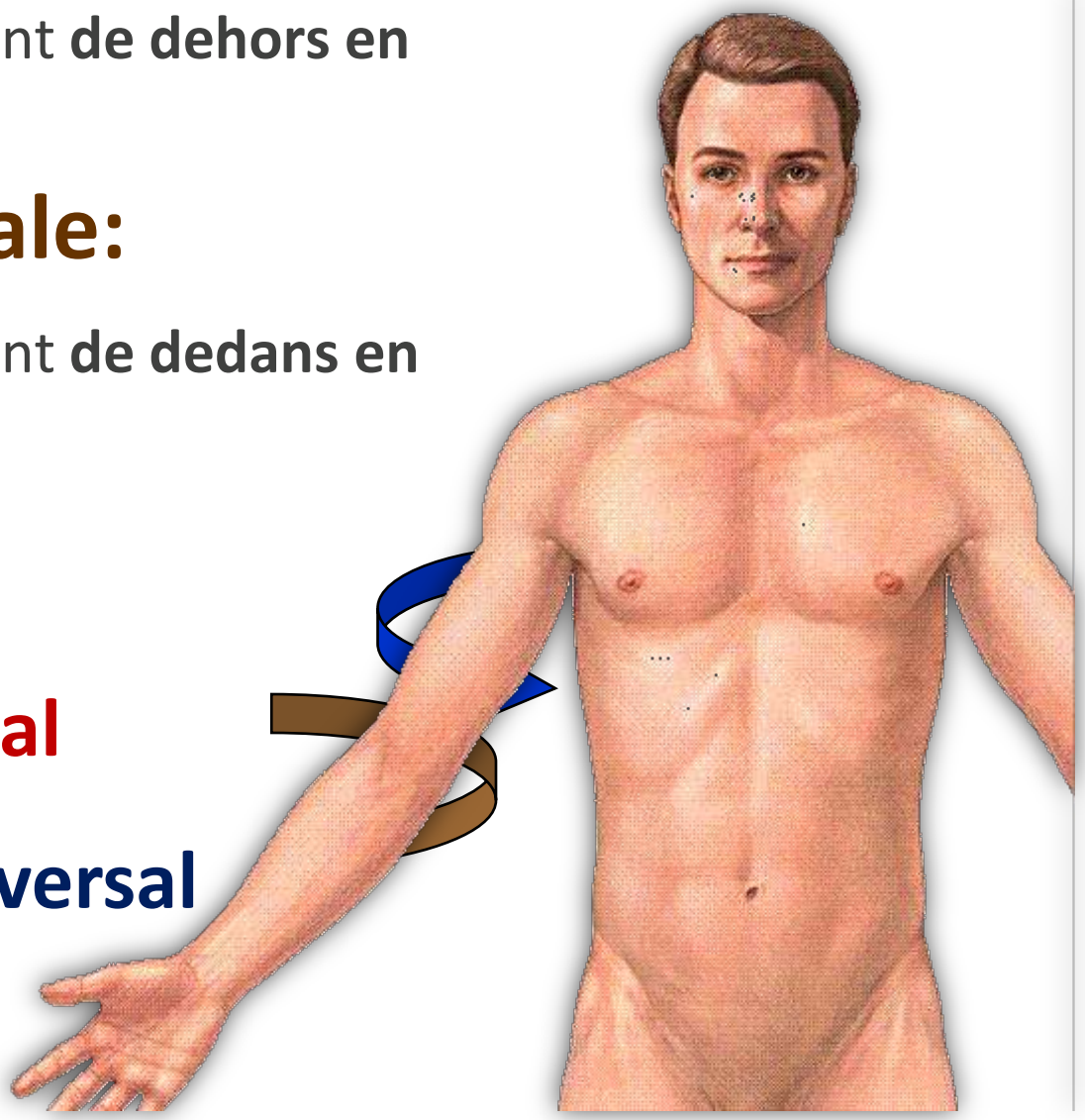
(4) Rotation latérale:

mouvement **rotatoire** en allant **de dedans en dehors**

Se font :

Selon un **axe vertical**

Dans un **plan transversal**



(5) Abduction :

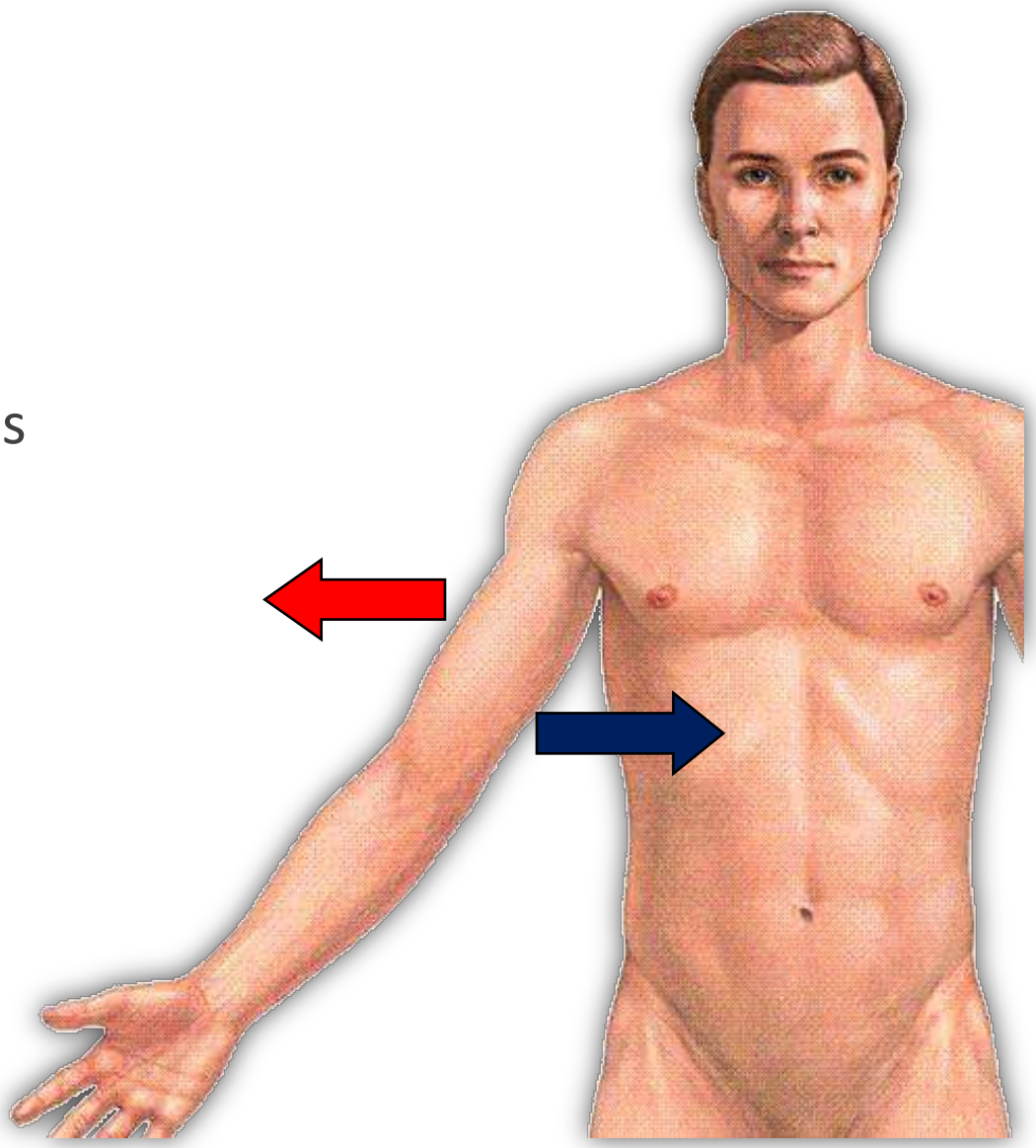
écarte une partie du corps
du **plan médian**

(6) Adduction :

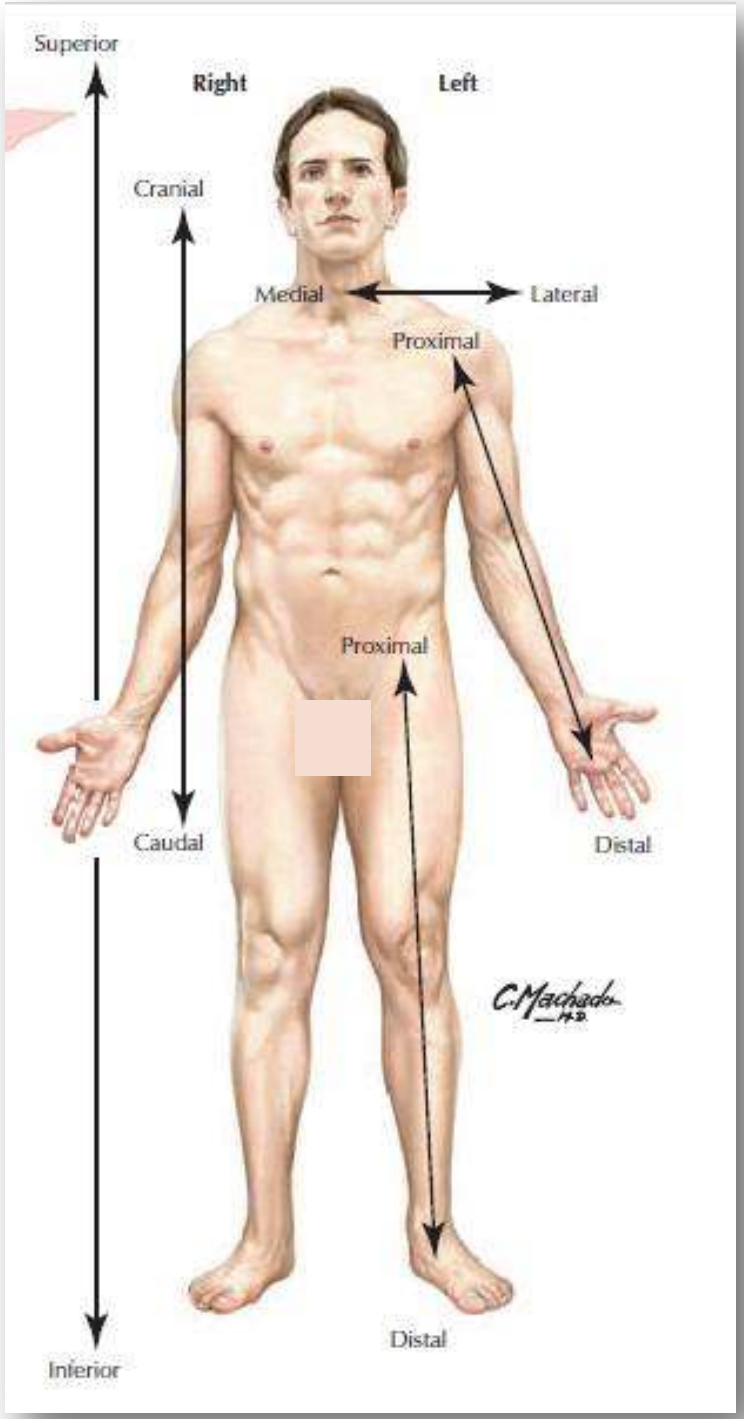
rapproche une partie du corps
au **plan médian**

Se font :

- Selon un **axe sagittal**
- Dans un **plan frontal**



Termes de localisation



Droit:
plus proche de la droite

Gauche:
plus proche de la gauche

Médial:
près du plan sagittal médian

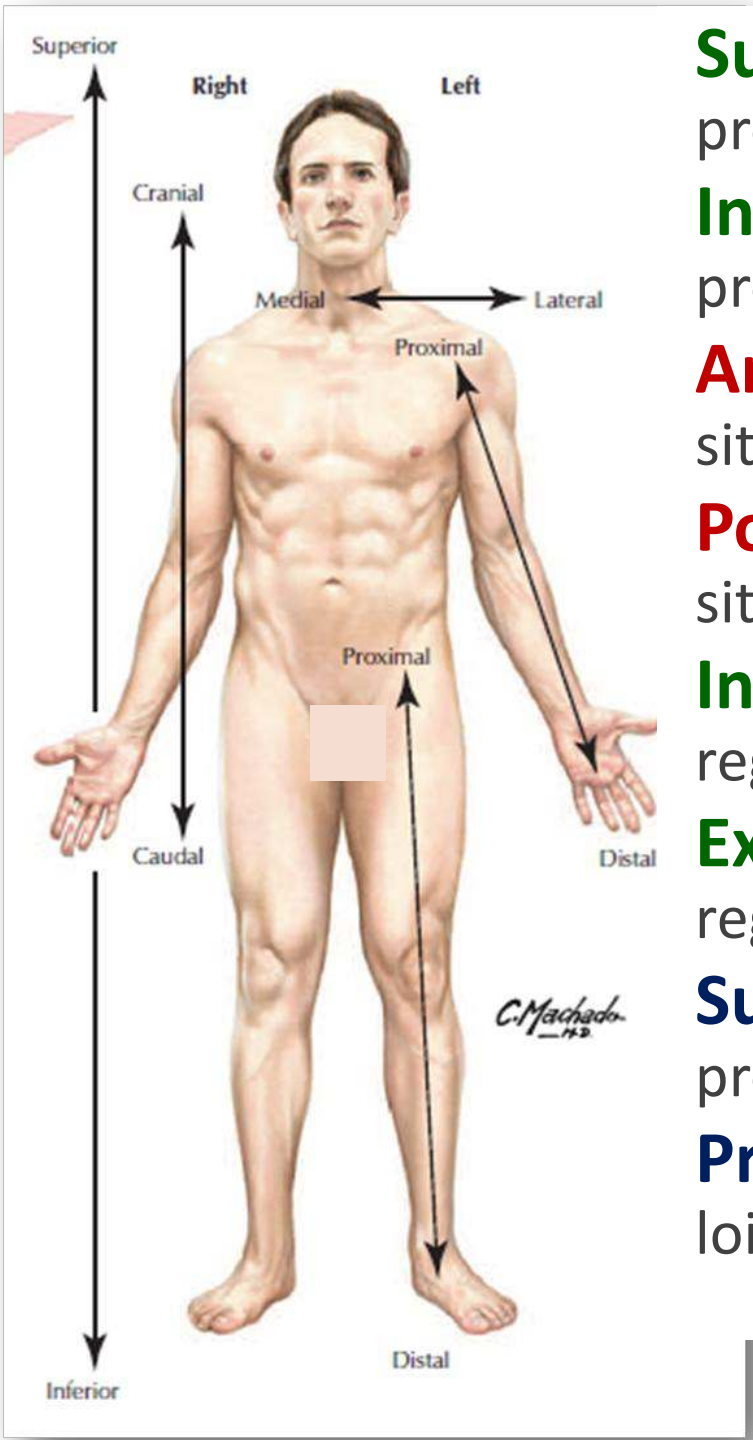
Latéral:
loin du plan sagittal médian

Proximal:
proche de la racine du membre

Distal:
loin de la racine du membre

Homolatéraux:
appartiennent au même côté

Controlatéraux:
appartiennent aux 2 côtés différents



Supérieur (**crânial** ou **céphalique**):
proche de l'extrémité supérieure

Inférieur (**caudal**):
proche de l'extrémité inférieure

Antérieur (**ventral**):
situé prés de la paroi antérieure

Postérieur (**dorsal**):
situé prés de la paroi postérieure

Interne:
regarde l'intérieur d'une cavité ou d'un viscère

Externe:
regarde l'extérieur d'une cavité ou d'un viscère

Superficiel:
proche de la surface

Profond:
loin de la surface

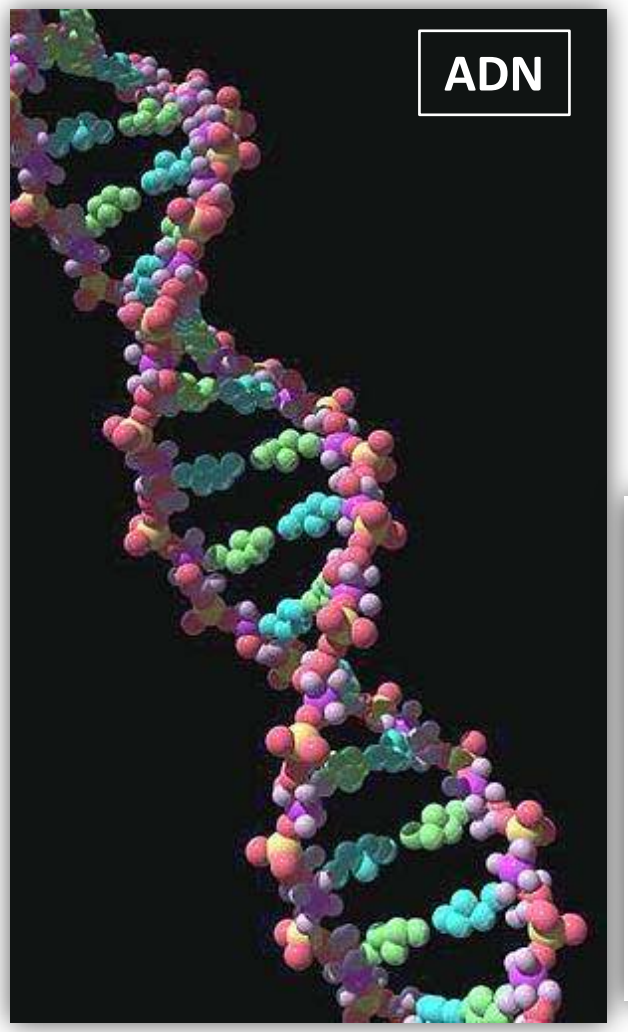


Organisation structurale du corps humain

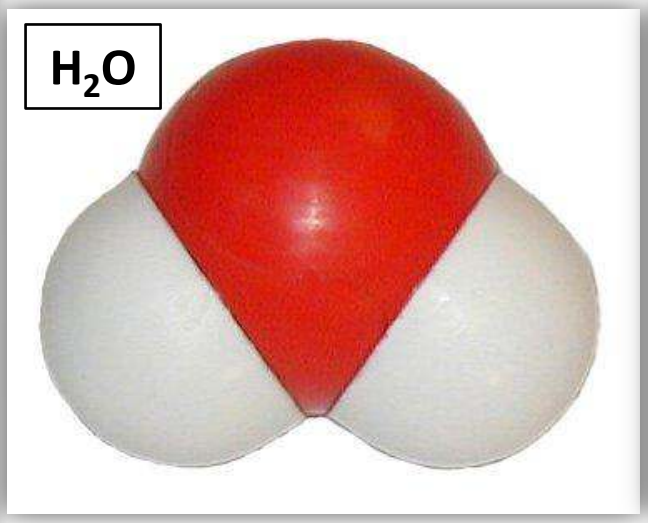


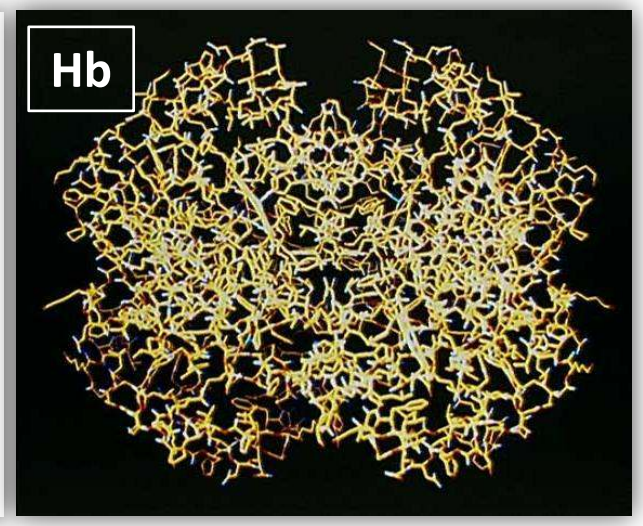
Niveaux d’organisation structurale

(1) Niveau chimique



L'ensemble des **substances chimiques** nécessaires au maintien de la structure et la vie

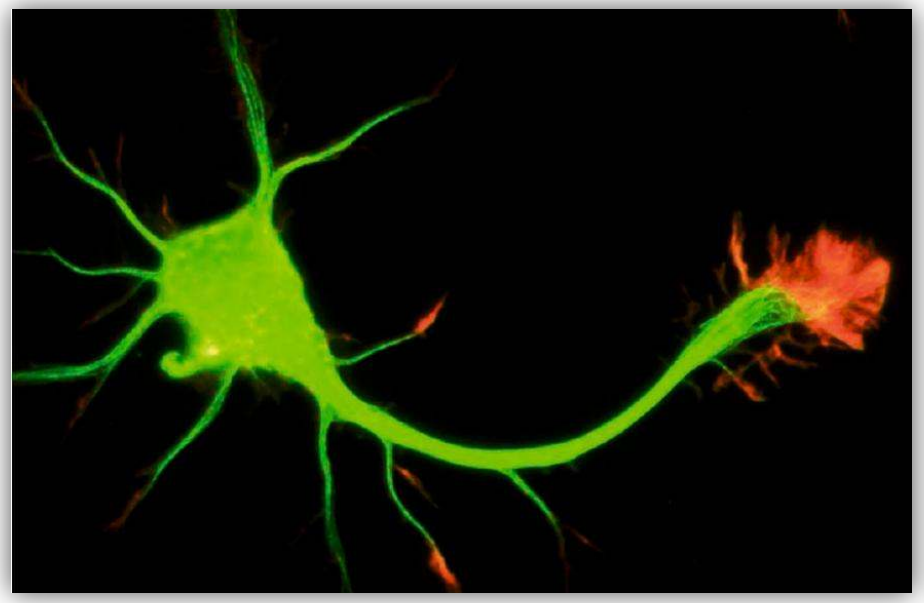




(2) Niveau cellulaire

La cellule représente l'unité structurale et **fonctionnelle de base**

Ex: neurone, cellule sanguine





(3) Niveau tissulaire

Le tissu représente le groupe

de cellules qui jouent un rôle

homologue

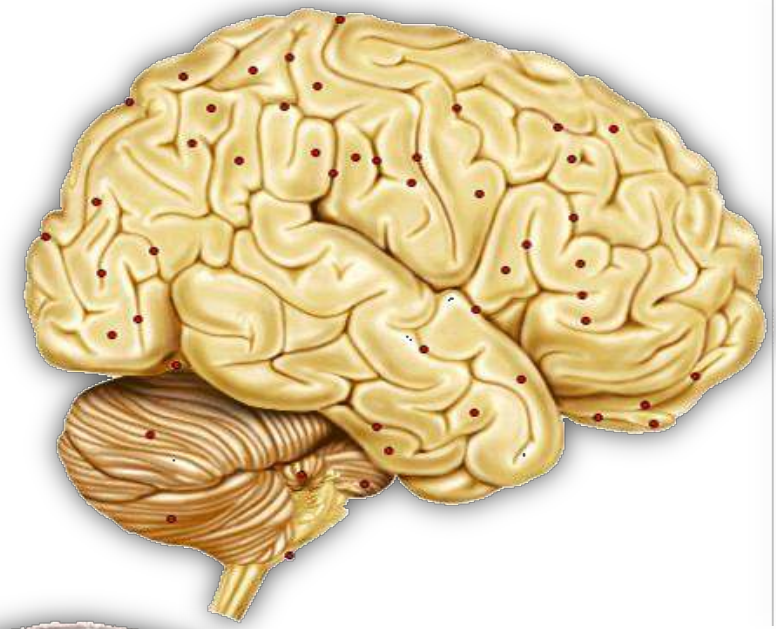
Ex: tissu osseux

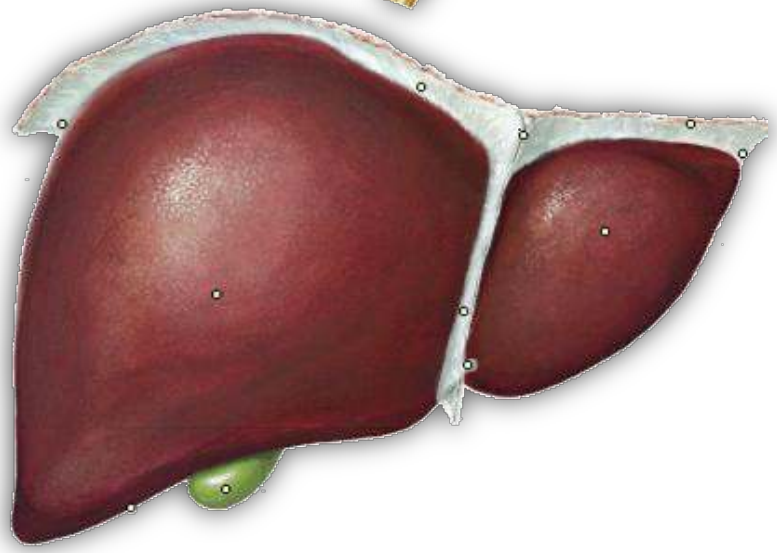


(4) Niveau organique

L'organe représente des tissus différents en vue d'une fonction spécifique

Ex: cerveau, foie

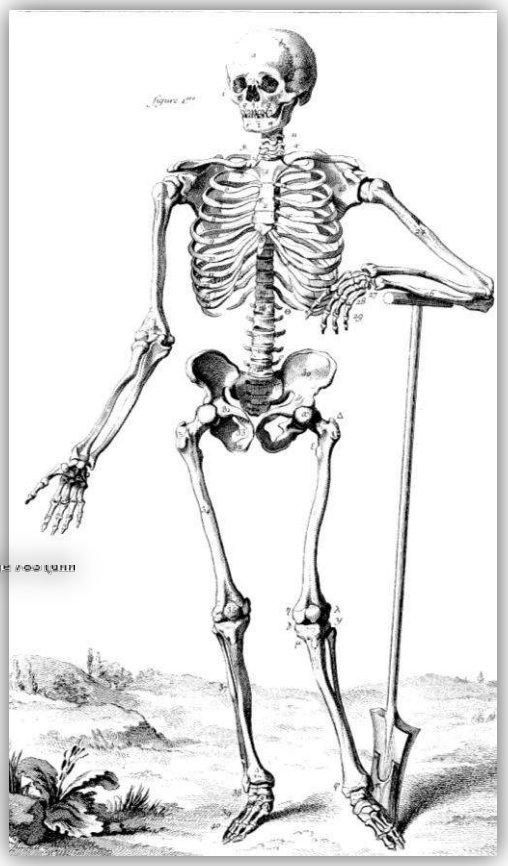


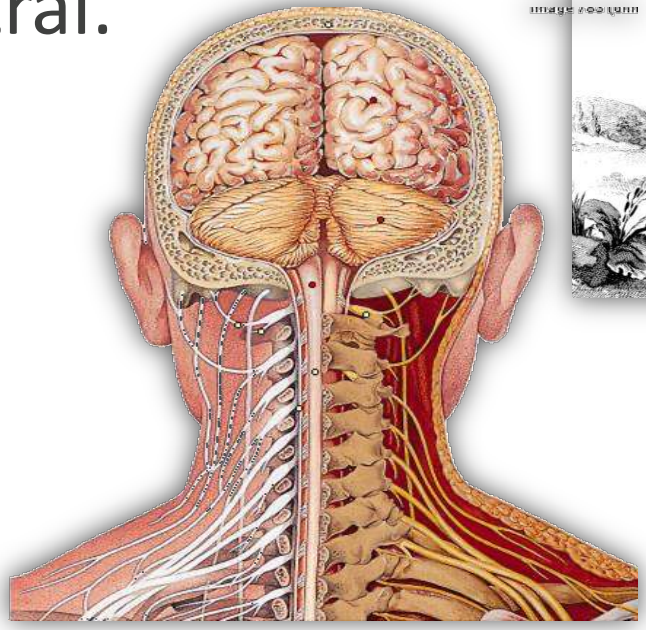


(5) Niveau systémique

Le système représente un ensemble d'organes de structure semblable.

Ex: système squelettique, système nerveux central.

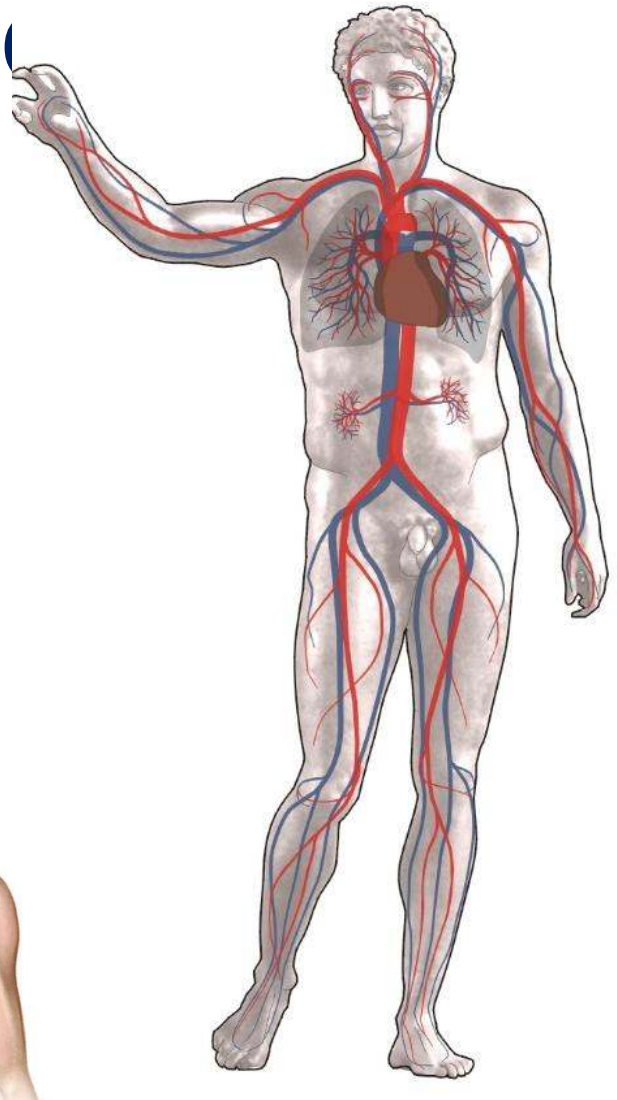


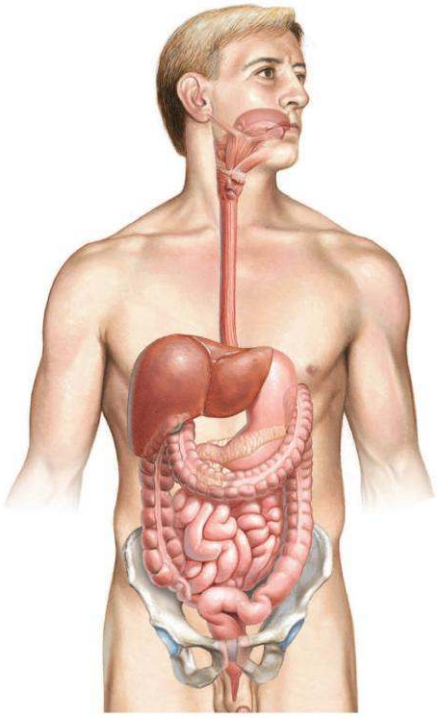


(6) Niveau systémiqu

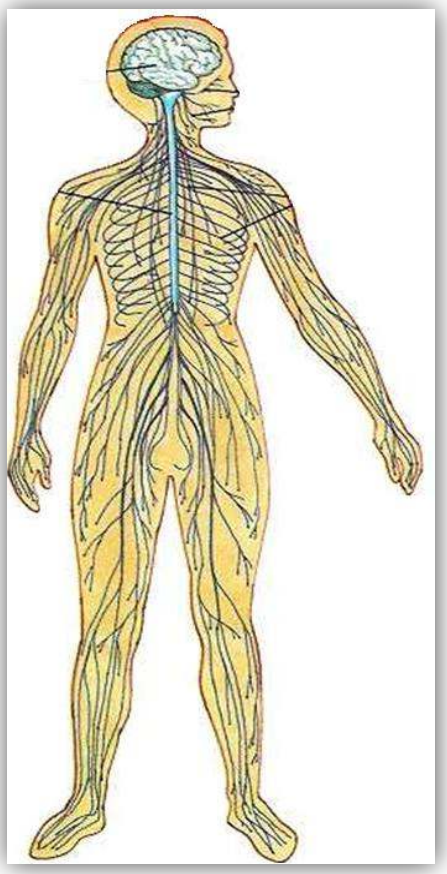
L'appareil représente un ensemble d'organes dissemblables, qui participent à une même fonction.

Ex: appareil digestif, appareil circulatoire.





Appareils du corps humain



Appareils de la vie de relation



Appareil locomoteur :

regroupe **systèmes osseux**, **articulaire** et **musculaire**

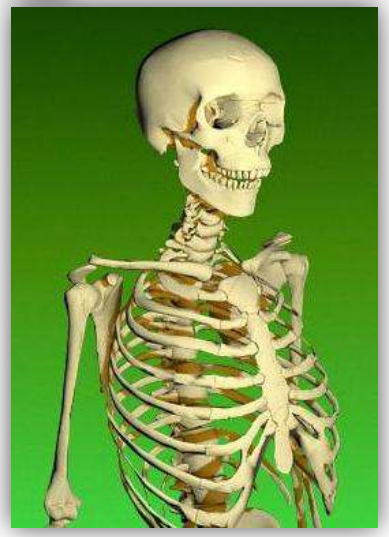
Appareil nerveux :

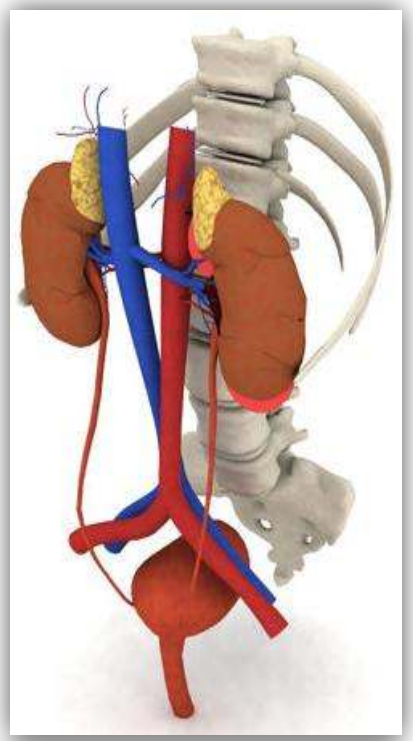
regroupe **systèmes nerveux central**, **périphérique** et **végétatif**

Appareil sensoriel :

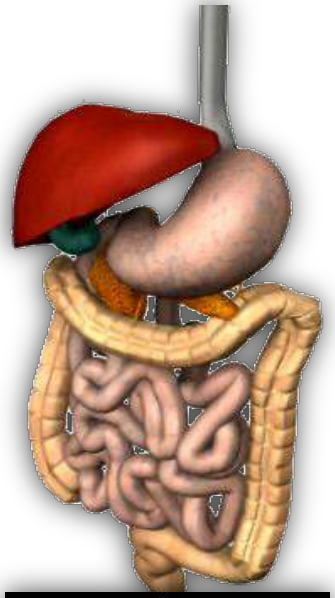
regroupe **5 organes de sens** (**toucher**, **goût**, **olfaction**, **vue** et **ouïe**).







Appareils de nutrition

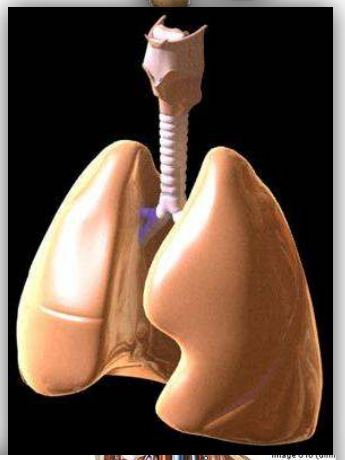


Appareil digestif :

Regroupe le **tube digestif** et les **glandes annexes**

Appareil urinaire :

Regroupe les **reins** et les **voies urinaires**

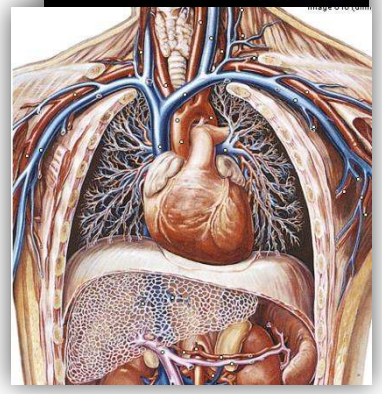


Appareil respiratoire :

Regroupe les **voies respiratoires** et les **poumons**

Appareil circulatoire :

Regroupe le **cœur** et le **systèmes artériel**, **veineux** et **lymphatique**



Système endocrinien :

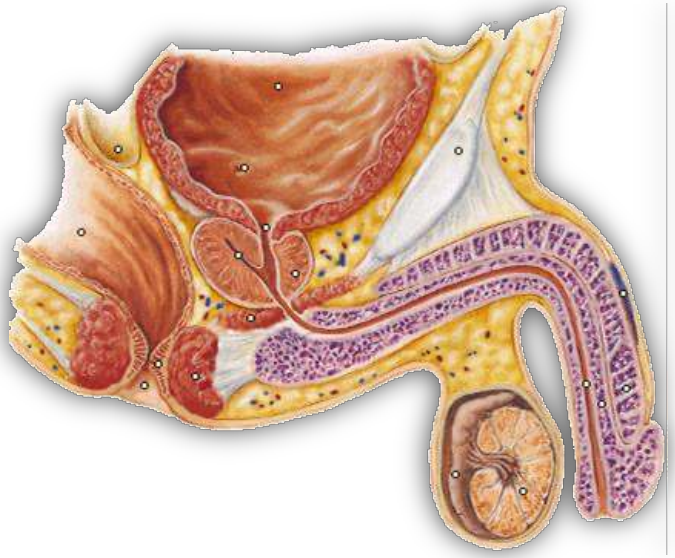
représenté par les **glandes endocrines**

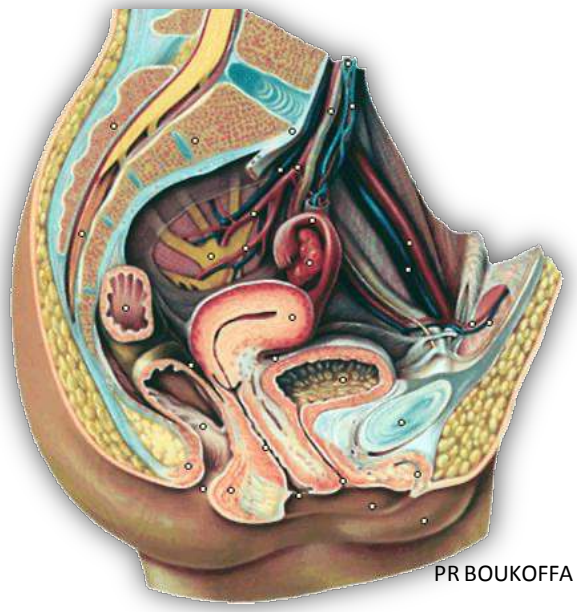


Appareil de la génération (de reproduction)

Appareil génital masculin:

regroupe: **testicules**, **voies spermatiques**, **verge** et **glandes annexes**





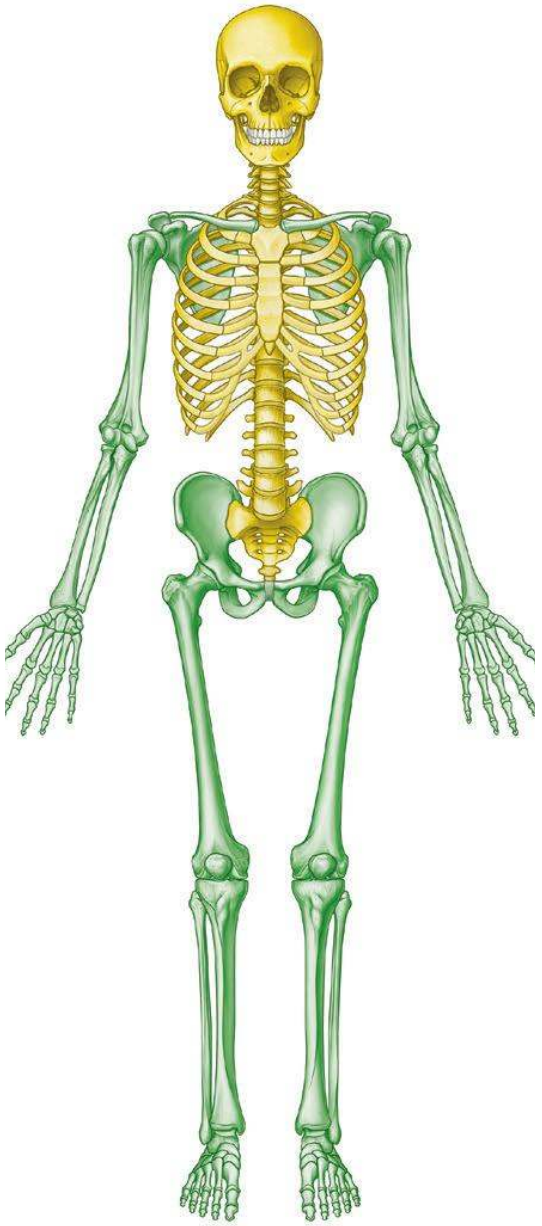
- **Appareil génital féminin:**

  regroupe: **utérus**, **annexes**, **vagin** et **vulve**

Organisation topographique du corps humain

Forme du corps humain

Déterminée par celle du **squelette ostéo-cartilagineux.**



Les muscles s'y ajoutent pour **remplissage** et **renforcement des parois.**



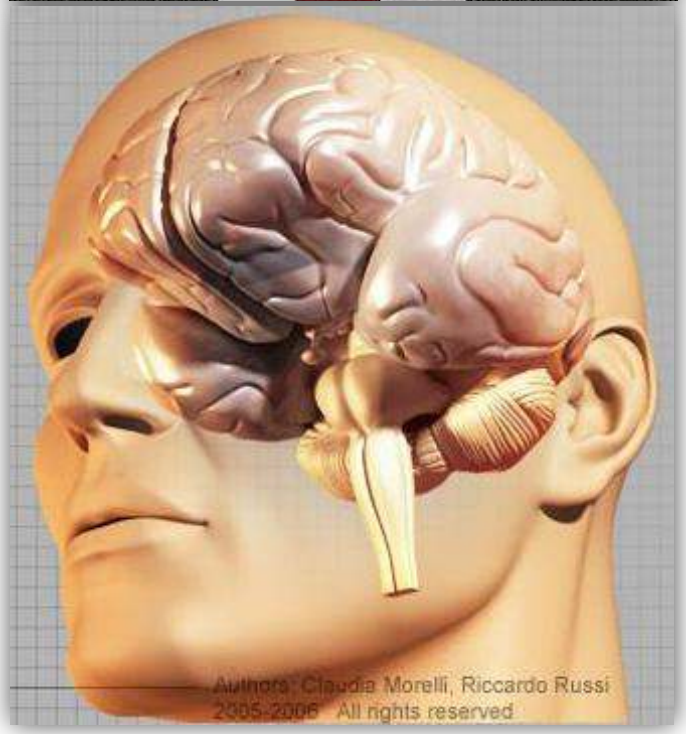
Tête

Emporte **l'appareil sensoriel.**

Contient **les segments initiaux** des **appareils digestif** et **respiratoire.**

Renferme la grande partie du **système nerveux.**



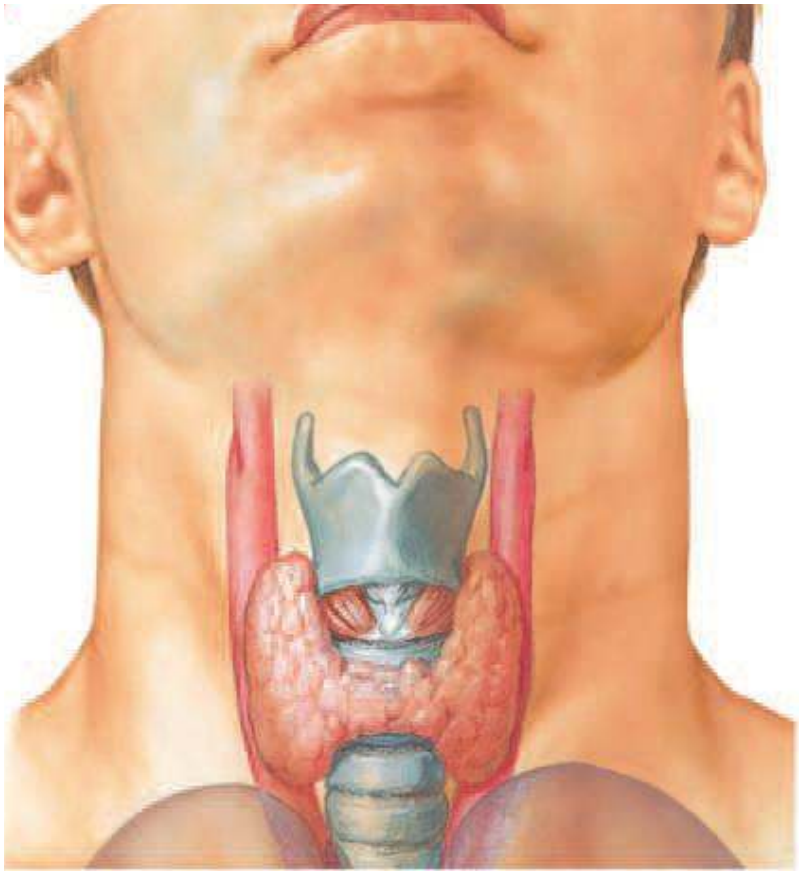


Cou

Supporte la tête.

Région de passage

pour des **éléments viscéraux**, **vasculaires** et **nerveux.**



Tronc



Renferme les **grandes cavités** du corps:

- En haut, **cavité thoracique**
- En bas, **cavité abdomino-pelvienne**



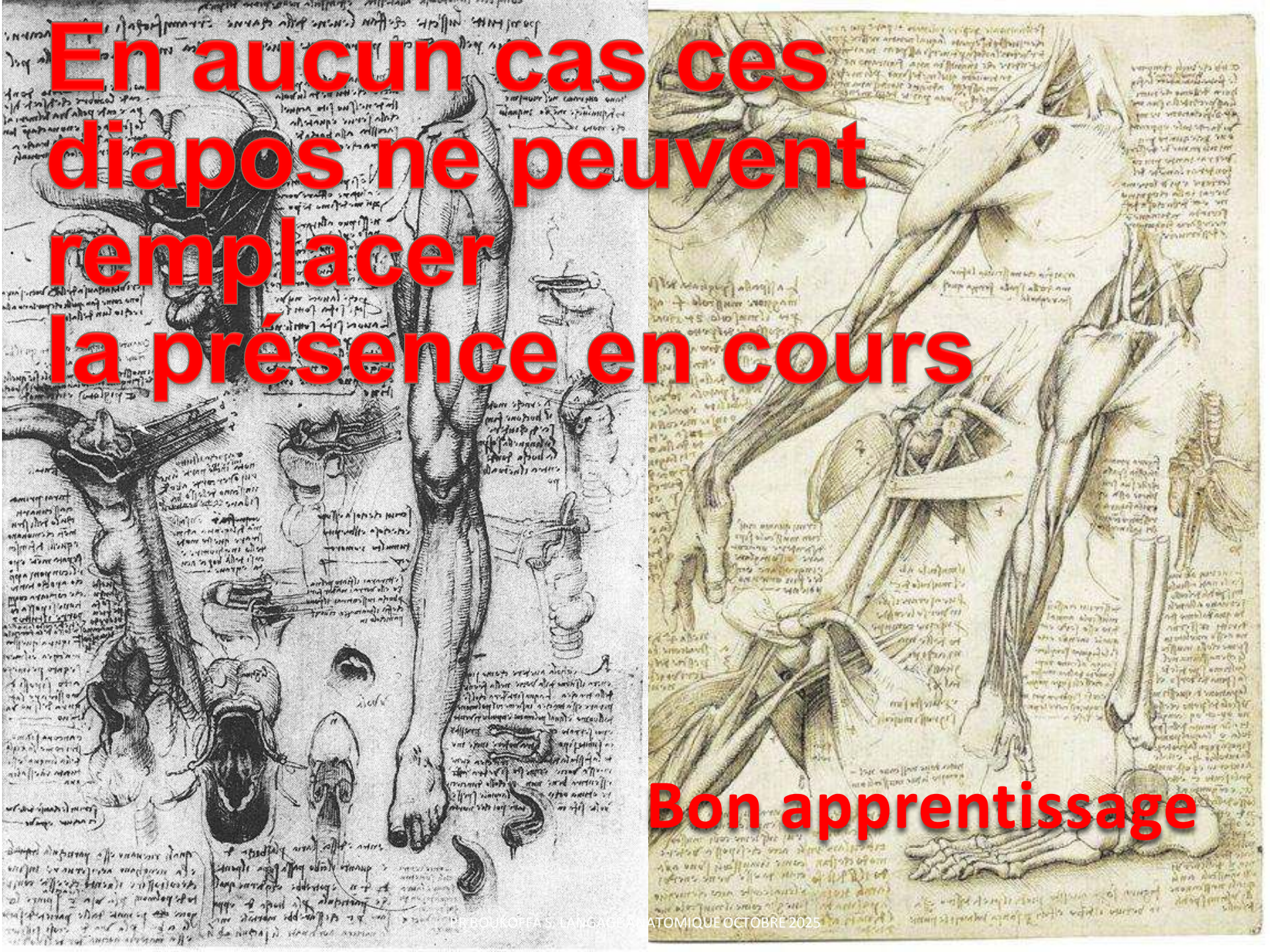
Membres

Extensions du tronc

Formés par:

- **Os**
- **Articulations**
- **Muscles**
- **Vaisseaux**
- **Nerfs**




En aucun cas ces diapos ne peuvent remplacer la présence en cours
Bon apprentissage